

PRIVACY RISKS IN AMAZON REVIEW DATA

RE-IDENTIFICATION THREATS IN ENTERPRISE DATA MONETIZATION

RELEASING BEHAVIOURAL DATASETS WITHOUT SAFEGUARDS IS LIKE PUBLISHING FINGERPRINTS WITHOUT NAMES... IDENTITIES CAN STILL BE RECOVERED.

SUMMARY

Mzansi Review Consulting was tasked with assessing re-identification risks in Amazon's product review dataset within a data monetization context. The dataset contains 21,054 reviews from 18,374 reviewers across 148 countries between 2007 and 2024. Our analysis shows that 12.4% of reviewers become uniquely identifiable when combining five quasi-identifiers: country, rating, day of week, posting time, and review length. A simulated linkage attack demonstrates that an adversary with access to four quasi-identifiers can correctly re-identify 38% of reviewers with zero false matches. To address this risk, three privacy-preserving strategies were evaluated: **generalization**, **suppression**, and **differential privacy**. A layered mitigation approach combining these techniques eliminates all uniquely identifiable records, achieves $k=5$ anonymity, and retains 99% of the dataset's analytical utility. The proposed privacy-by-design architecture demonstrates that ethical data governance can protect individuals while supporting sustainable enterprise data monetization.

KEY FINDINGS

- 12.4% of records uniquely identifiable using five quasi-identifiers
- 38.2% re-identification success rate using four quasi-identifiers
- Geographic and temporal attributes contain the highest identifying entropy
- k -anonymity mitigation ($k \geq 5$) removes re-identification risk while preserving data utility
- Layered mitigation reduces uniqueness from 12.4% to 0%

ETHICAL INSIGHT

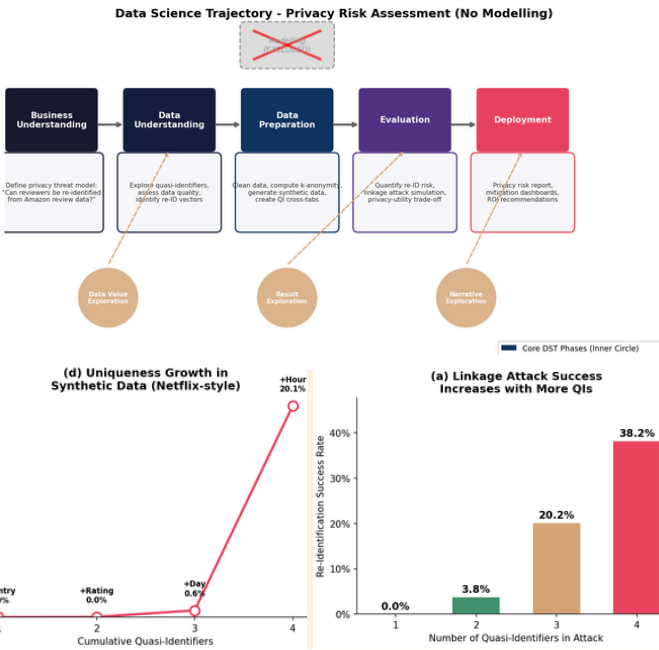
Removing direct identifiers does not guarantee anonymity. Quasi-identifiers such as location, timestamps, and behavioural patterns create behavioural fingerprints. When combined with auxiliary datasets, these attributes allow attackers to reconstruct identities through linkage attacks. Responsible data science therefore requires privacy-by-design principles, ensuring that enterprises apply statistical privacy safeguards before sharing or monetising user data.

METHODOLOGY & CHARTS

DATA MANAGEMENT TRAJECTORY

Procedures:

- Data Release Analysis
 - Evaluated risks associated with direct dataset release compared to privacy-preserving data abstraction approaches.
- Entropy & Uniqueness Analysis
 - Computed Shannon entropy and k -anonymity levels to quantify the identifying power of quasi-identifiers.
- Privacy Mitigation
 - Applied generalisation and suppression ($k \geq 5$) to reduce uniqueness and enforce k -anonymity.
- Statistical Privacy Protection
 - Implemented differential privacy to protect aggregate outputs while preserving analytical utility.



CONCLUSION

As enterprises increasingly pursue data monetization strategies, the risk of unintentionally exposing individuals through behavioural datasets grows significantly. This study demonstrates that even anonymized review data can reveal identities when quasi-identifiers are combined. However, a layered privacy architecture combining generalization, suppression, and differential privacy effectively eliminates re-identification risk while preserving analytical value. The findings highlight that privacy-preserving data sharing is both an ethical obligation and a strategically sound business decision for organisations operating data-driven platforms